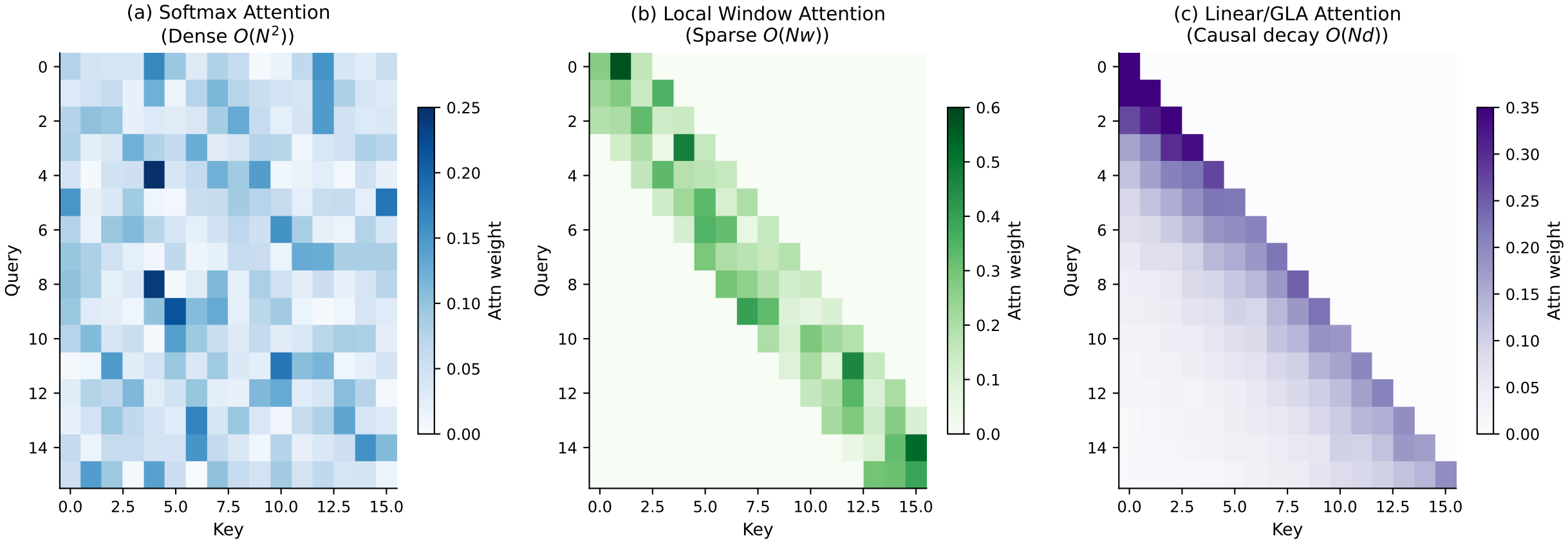
Learned Attention Patterns: Dense vs. Sparse vs. Linear Causal
(a) Softmax Attention
(Dense $O(N^2)$)
(b) Local Window Attention
(Sparse $O(Nw)$)
(c) Linear/GLA Attention
(Causal decay $O(Nd)$)
Query
Key
Attn weight
0
2
4
6
8
10
12
14
0.0
2.5
5.0
7.5
10.0
12.5
15.0
0.25
0.20
0.15
0.10
0.05
0.00
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00